

Elementary Problems

Double Counting

Solutions

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Basic Work

1. We count the number of pairs (S, C) where S is a student and C is a club for which S joins. For each student S , there are 3 pairs for which the first entry is S . So there are $3s$ pairs where s is the number of students. For each club C , there are 3 pairs for which the second entry is C . So there are $3c$ pairs where c is the number of clubs. Therefore, we get $3s = 3c$, which yields $s = c$.
2. There are $\binom{n}{2} = \frac{n(n-1)}{2}$ pairs of elements. Each pair appears in at most one subset. Also, each subset A_j contains $\binom{3}{2} = 3$ pairs of elements. This implies $3m \leq \frac{n(n-1)}{2}$, and hence $m \leq \frac{n(n-1)}{6}$.
3. Suppose on the contrary that no such pair of students exists. We count the number N of pairs $(P, \{S_1, S_2\})$ such that both students S_1 and S_2 cannot solve the problem P . For each problem P , there are 3 students not solving this problem. So there are $\binom{3}{2} = 3$ pairs with P as the first entry. This implies $N = 3 \times 8 = 24$.
For each pair of students S_1 and S_2 , there exists a problem P such that $(P, \{S_1, S_2\})$ is such a pair by the assumption. So we have $N \geq \binom{8}{2} = 28$, which is a contradiction. Hence, there are 2 students such that every problem is solved by at least one of them.
4. Suppose there are n_k classes with exactly k students. Then any two of these classes have at most one common student. Note that there are $\binom{k}{2}$ pairs of students in each of these classes. Therefore, the condition implies

$$n_k \binom{k}{2} \leq \binom{n}{2},$$

which is the total number of pairs of students. This yields

$$n_k \leq \frac{n(n-1)}{k(k-1)}.$$

Summing over k from 2 to n , the total number of classes is at most

$$\begin{aligned} & \frac{n(n-1)}{1 \times 2} + \frac{n(n-1)}{2 \times 3} + \cdots + \frac{n(n-1)}{n(n-1)} \\ &= n(n-1) \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{n-1} - \frac{1}{n} \right) \\ &= n(n-1) \left(1 - \frac{1}{n} \right) = (n-1)^2. \end{aligned}$$

5. For each question in the test, we claim that there are at least $n + 1$ students who get correct answers. Indeed, if there are at most n students answering this question correctly, then they could form a group (possibly with some other students) so that the other group does not contain any student who gets a correct answer in this question. This contradicts the given fact.

Suppose there are k questions in total. From above, there are at least $k(n + 1)$ correct answers in total. On average, each student gets at least $\frac{k(n + 1)}{2n} > \frac{k}{2}$ correct answers. Hence, there must be a student who gets more than half of the questions correctly.

6. We use numbers to denote the heights of the children. Suppose the numbers in the first row are $a_1 < a_2 < \dots < a_n$ and the numbers in the second row are $b_1 < b_2 < \dots < b_n$. We need to show that $b_j > a_j$ for each j .

Consider any b_j . It is larger than the numbers $b_1, \dots, b_{j-1}$ in the second row. Each of the numbers $b_1, b_2, \dots, b_j$ is larger than a distinct number in the first row, which is the number in front of it initially. Therefore, b_j is larger than at least j numbers in the first row. It follows from the order that $b_j > a_j$. This completes the proof.

7. Suppose such an alignment is possible. Note that there are 7 halves of the dominoes each of which has 1 dot. Indeed, they occur in the pairs $(1, 1), (1, 2), \dots, (1, 6)$. These halves can be paired up such that each pair is adjacent in the cycle. It is impossible to have such a pairing since 7 is odd. So it is not possible to align the dominoes in the desired way.
8. Let b_j and w_j be the sum of numbers in the black cells and the white cells in the j th row respectively. Since $b_j + w_j$ is even, b_j and w_j have the same parity. Consider the sum of all numbers in the even columns. On the one hand, as the sum of each column is even, this sum is even. On the other hand, this is equal to $w_1 + b_2 + w_3 + b_4 + \dots$ if the top left corner cell is black, and is equal to $b_1 + w_2 + b_3 + w_4 + \dots$ if not. In either case, this has the same parity as $b_1 + b_2 + b_3 + \dots$, which is the sum of all numbers in the black cells. Thus, the sum of all numbers in the black cells is even.
9. Suppose such an arrangement is possible. Consider a particular person A . Let k be the number of times that A is on duty. During each duty, A patrols with exactly 2 other people. Thus, A patrols with exactly $2k$ people in total. This contradicts the fact that there are 99 people other than A . Thus, it is not possible to make every pair of people to be on duty simultaneously for exactly one time.

10. Yes. Although one simply needs to give a possible construction, we explain how to obtain such a construction in details. The idea also works if such a construction does not exist.

Label the ports $1, 2, 3, 4, 5, 6$. Since each of the $\binom{6}{2} = 15$ pairs belong to exactly

two routes, and each route consists of $\binom{3}{2} = 3$ pairs, the number of routes is exactly $\frac{15 \times 2}{3} = 10$.

For each port, say port 1, if it belongs to exactly m routes, then there are $2m$ pairs containing port 1 counted with multiplicity since there are 2 other ports in each of the m routes. Note that there are 5 pairs of ports containing port 1 and each of them belongs to exactly 2 routes. This gives $2m = 5 \times 2$ and hence $m = 5$. This means each port belongs to exactly 5 ports.

These are what we need. There are 5 routes containing port 1.

$$(1XX, 1XX, 1XX, 1XX, 1XX, XXX, XXX, XXX, XXX, XXX)$$

As the pair $(1, 2)$ appears twice, 2 of these routes contain 1, 2, and there are 3 additional routes containing port 2 but not 1. WLOG the first two routes are 123 and 124.

$$(123, 124, 1XX, 1XX, 1XX, 2XX, 2XX, 2XX, XXX, XXX)$$

This also shows that for any two labels, exactly 8 of the routes consists of at least one of these labels. Thus, there cannot be a route 134 or 234, since otherwise there are insufficient routes for ports 5 and 6. Noting that each of the pairs $(1, 3)$, $(1, 4)$, $(2, 3)$, $(2, 4)$ appears twice, we can work out some more routes up to symmetry.

$$(123, 124, 13X, 14X, 1XX, 23X, 24X, 2XX, 34X, 34X)$$

It is not hard to complete all the routes as follows.

$$(123, 124, 135, 146, 156, 236, 245, 256, 345, 346)$$

One checks that the conditions are satisfied.

11. Suppose it is possible to write down the numbers. There are 12 black faces, each of which is adjacent to 5 white faces. Let B and W be the sums of numbers on the black faces and the white faces respectively. From the condition, B equals the sum of numbers on $5 \times 12 = 60$ white faces, where each white face of the football is counted 3 times as each white face is adjacent to 3 black faces. This implies $B = 3W$.

As $B + W = 1 + 2 + \dots + 32 = 528$, we find that $4W = 528$, and hence $W = 132$. However, this is impossible since $W \geq 1 + 2 + \dots + 20 = 210$, as there are 20 white faces.

Therefore, it is not possible that the condition is satisfied.

12. Note that there are 2^n subsets and hence 2^{nm} tuples of subsets. So

$$\sum_{(A_1, A_2, \dots, A_m)} |A_1 \cup A_2 \cup \dots \cup A_m| = n \cdot 2^{nm} - \sum_{(A_1, A_2, \dots, A_m)} (n - |A_1 \cup A_2 \cup \dots \cup A_m|).$$

The sum on the right-hand side counts the number N of tuples $(A_1, A_2, \dots, A_m, k)$ such that each A_j is a subset of $\{1, 2, \dots, n\}$ not containing k and $1 \leq k \leq n$, since each tuple $(A_1, A_2, \dots, A_m)$ contributes $n - |A_1 \cup A_2 \cup \dots \cup A_m|$ to this number.

For each $1 \leq k \leq n$, there are 2^{n-1} subsets not containing k . So k contributes $2^{(n-1)m}$ to N . It follows that the desired quantity is

$$n \cdot 2^{nm} - N = n \cdot 2^{nm} - n \cdot 2^{(n-1)m} = n \cdot 2^{(n-1)m}(2^m - 1).$$

13. We count the number of pairs (m, σ) where $1 \leq m \leq n$ and m is a fixed point under the permutation σ . Firstly, for each fixed m , we can permute any other $n - 1$ numbers arbitrarily to form a permutation σ . So there are $(n - 1)!$ pairs with m as the first entry. Hence, there are $n \cdot (n - 1)! = n!$ pairs.

On the other hand, for each $0 \leq k \leq n$, we count the number of pairs with σ being a permutation with exactly k fixed points. By definition, there are $p_n(k)$ such permutations σ . For each σ , there are k choices of m which are fixed points. Thus, the number of pairs is $\sum_{k=0}^n k \cdot p_n(k)$. It follows that

$$\sum_{k=0}^n k \cdot p_n(k) = n!.$$

14. If n is even, we can remove any row and any column. Since the remaining board has an odd number $(n - 1)^2$ of cells, the number of black cells and the number of white cells must be different.

If n is odd, WLOG we may assume there are more black cells. Let $b \geq \frac{n^2 + 1}{2}$ be the number of black cells. For each pair (i, j) of indices, let $a_{i,j}$ be the number of black cells remaining after removing the i th row and the j th column. Consider the sum

$$\sum_{1 \leq i, j \leq n} a_{i,j}.$$

Each black cell is counted $(n - 1)^2$ times since there are $(n - 1)^2$ pairs of rows and columns which do not contain this cell. Therefore, we have

$$\sum_{1 \leq i, j \leq n} a_{i,j} = (n - 1)^2 b \geq \frac{(n - 1)^2(n^2 + 1)}{2}.$$

By the pigeonhole principle, there exists $a_{i,j}$ such that

$$a_{i,j} \geq \left\lceil \frac{(n - 1)^2(n^2 + 1)}{2n^2} \right\rceil = \frac{(n - 1)^2}{2} + \left\lceil \frac{(n - 1)^2}{2n^2} \right\rceil = \frac{(n - 1)^2}{2} + 1.$$

Hence, by deleting the i th row and the j th column, the remaining board has more black cells than white cells.

15. Fix any team A . Let $B_1, B_2, \dots, B_d$ be all the teams won by A . There are $\binom{d}{2}$ matches played among the teams $B_1, B_2, \dots, B_d$, resulting in $\frac{d(d-1)}{2}$ pairs (B_i, B_j) such that B_i wins over B_j .

On the other hand, we count the number of pairs (B_i, B_j) such that B_i wins over B_j for fixed $1 \leq i \leq d$. Note that the pair A and B_i win over exactly t teams, and these t teams must be some of the B_j 's since these are the only teams that lose to A . Thus, each B_i corresponds to exactly t pairs (B_i, B_j) . By considering all i 's, there are exactly dt pairs.

Thus, we obtain $\frac{d(d-1)}{2} = dt$, so that $d = 2t + 1$. This holds for any team A . In other words, every team wins over exactly $2t + 1$ other teams.

Similarly, we count the number of matches played in two ways. Firstly, there are obviously $\binom{n}{2} = \frac{n(n-1)}{2}$ matches. Secondly, each of the n teams wins exactly $2t + 1$ times. Thus, there are $n(2t + 1)$ matches in total. From

$$\frac{n(n-1)}{2} = n(2t + 1),$$

we have $n = 2(2t + 1) + 1 = 4t + 3$.

16. Consider the number N of pairs (A, B) of subsets of T such that for any $a \in A$ and $b \in B$, we have $(a, b) = 1$. For $A \neq B$, the pair (A, B) is counted if and only if (B, A) is counted. For $A = B$, the pair (A, B) is counted if and only if it is the trivial case $A = B = \emptyset$. Therefore, N is odd.

For each good subset A of T , the pair (A, B) is counted if $B = \emptyset$. If $B \neq \emptyset$, then we pick any $b \in B$, and we can always find $a \in A$ such that $(a, b) > 1$ as A is good. Thus, A contributes 1 to N .

For each subset A of T which is not good, suppose there are $k \geq 1$ elements t in T such that $(a, t) = 1$ for all $a \in A$. Then the pair (A, B) is counted if and only if B only contains some of these k elements. Thus, there are 2^k such pairs. As $k \geq 1$, the contribution of A to N is even. Note that this also holds when $A = \emptyset$.

From above, we find that N has the same parity as the number of good subsets of T . Hence, the number of good subsets is odd.

17. We use tasks I and II to denote working in the diamond mine and collecting berries respectively. Suppose on the contrary that there are at most 6 dwarves performing task II on each day. From the condition, by considering any day with the first day, there are at least 3 dwarves performing task II on each of the remaining days. Let a, b, c, d be the number of days for which exactly 3, 4, 5, 6 dwarves perform task II respectively. Then $a + b + c + d = 15$.

Firstly, if $d \geq 2$, then on any 2 of these days, there are at least $7 - 1 - 1 = 5$ dwarves performing task II on both days. This is a contradiction since there should be 3 dwarves working on different tasks. So $d \leq 1$.

Next, consider those days counted in b . If there are 2 days which share at least 2 dwarves performing task I on both days, then there are 5 dwarves performing the same task on both days. This is a contradiction. We count the number of pairs of dwarves both performing task I on these b days. Firstly, since there are 3 dwarves performing task I on each day, there are $\binom{3}{2}b = 3b$ pairs. All these pairs are distinct from the above observation. As there are 7 dwarves, the number of pairs is at most $\binom{7}{2} = 21$. This yields $3b \leq 21$, and hence $b \leq 7$.

If $d = 1$, then the only dwarf who performs task I on that day cannot perform task I on any day counted in b . So the number of pairs counted above is at most $\binom{6}{2} = 15$. This yields $3b \leq 15$, and hence $b \leq 5$. So we have

$$b + d \leq \max \{1 + 5, 0 + 7\} = 7.$$

Similarly, by interchanging the roles of tasks I and II, we find that $a \leq 7$. Also, if we consider those days counted in c , we find that there cannot be 2 days which share a common dwarf performing task I on both days. This leads to the bound $c \leq \left\lfloor \frac{7}{2} \right\rfloor = 3$.

If $c = 1$, suppose dwarves A, B perform task I and C, D, E, F, G perform task II on that day. Then on each day counted in a , at least one of A and B should perform task II. We count the number of pairs (X, Y) of dwarves both performing task II on the days counted in a where $X = A, B$ and $Y = C, D, E, F, G$. Note that each day contributes 2 such pairs. Also, there are at most $2 \times 5 = 10$ pairs. This yields $2a \leq 10$, and hence $a \leq 5$.

If $c = 2, 3$, suppose dwarves A, B perform task I on one of these days and C, D perform task I on another day. As above, on each day counted in a , at least one of A, B and at least one of C, D should perform task II. Since no pair can be repeated, we have $a \leq 2 \times 2 = 4$. It follows that

$$a + c \leq \max \{7 + 0, 5 + 1, 4 + 3\} = 7.$$

But this yields the contradiction $15 = (b + d) + (a + c) \leq 7 + 7 = 14$. Hence, there must be a day for which all dwarves are performing task II.

18. We consider the $(m - 1)(n - 1)$ intersection points of the grid lines in the interior of the board. So each point is the endpoint of exactly 4 edges of the unit squares. We say that an edge is good if it separates two consecutive numbers. For each point P , we count the number N_P of good edges having it as an endpoint.

Firstly, $N_P \leq 3$. Indeed, if $N_P = 4$ and the numbers surrounding P are a, b, c, d in the clockwise direction, then we must have $b = a \pm 1$, $c = a \pm 2$, $d = a \pm 3$ and hence the contradiction $a = a \pm 4$. If $N_P = 3$, suppose the numbers surrounding P are a, b, c, d in the clockwise direction such that a and d are not separated by a good edge. As above, we have $d = a \pm 3$. So it remains to show there exists a point P with $N_P = 3$.

5	4	3	2	1
6	13	14	15	16
7	12	11	18	17
8	9	10	19	20
25	24	23	22	21

Let $S = \sum_P N_P$. Every good edge separating two squares on the boundary of the board contributes 1 to S , since one of its endpoints is not under consideration. Every other good edge contributes 2 to S . There are $2(m-1) + 2(n-1)$ edges on the boundary of the board. However, it is not possible that all of them are good edges, or otherwise we get a cycle of strictly increasing numbers either in the clockwise or anticlockwise direction. Therefore, at most $2m + 2n - 5$ good edges contribute 1 to S . As there are $mn - 1$ good edges, this yields

$$S \geq 2(mn - 1) - (2m + 2n - 5) = 2(m-1)(n-1) + 1.$$

By the pigeonhole principle, there exists P such that

$$N_P \geq \left\lceil \frac{2(m-1)(n-1) + 1}{(m-1)(n-1)} \right\rceil = 3.$$

As $N_P \leq 3$, we must have $N_P = 3$. We are done as explained above.

19. Clearly, none of the subsets is empty. If one subset contains all n elements, then every other subset contains exactly 1 element. Since they are distinct, we can only have 2 subsets, in which case $m \leq n$ holds trivially. If all subsets contain the same element, then every other element belongs to at most 1 subset. It follows easily that $m \leq n$ since otherwise there are at least n elements from 2 to n .

Now, we may assume all subsets are nonempty and contain at most $n-1$ elements, and no element belongs to all subsets. Let N be the number of pairs (S_j, k) such that

$1 \leq j \leq m$, $1 \leq k \leq n$ and $k \in S_j$. Firstly, each subset S_j belongs to $|S_j|$ pairs. So we have

$$N = \sum_{j=1}^m |S_j|.$$

Define $f(S_j, k) = 1$ if $k \in S_j$ and $f(S_j, k) = 0$ if $k \notin S_j$. We rewrite the above as follows.

$$N = \sum_{j=1}^m (n - |S_j|) \frac{|S_j|}{n - |S_j|} = \sum_{j=1}^m \sum_{k=1}^n (1 - f(S_j, k)) \frac{|S_j|}{n - |S_j|}. \quad (1)$$

Next, suppose the element k belongs to exactly a_k subsets. Then

$$N = \sum_{k=1}^n a_k = \sum_{k=1}^n (m - a_k) \frac{a_k}{m - a_k} = \sum_{k=1}^n \sum_{j=1}^m (1 - f(S_j, k)) \frac{a_k}{m - a_k}. \quad (2)$$

Comparing (1) and (2), we must have

$$\frac{|S_j|}{n - |S_j|} \leq \frac{a_k}{m - a_k}$$

for some $1 \leq j \leq m$ and $1 \leq k \leq n$ satisfying $k \notin S_j$. This implies

$$m|S_j| \leq na_k.$$

We now claim that $|S_j| \geq a_k$, from which it follows that $m \leq n$. Indeed, for every other subset S_i containing k , as $|S_i \cap S_j| = 1$ and $k \notin S_j$, there must be a unique element $t \in S_i \cap S_j$ distinct from k . Two distinct subsets containing k must correspond to distinct elements t since otherwise they would have 2 elements in common. Thus, the number of subsets containing k does not exceed the number of elements in S_j , i.e. $a_k \leq |S_j|$. The proof follow from the above discussion.

Binomial Coefficient

20. We count the number of subsets of $\{1, 2, \dots, n\}$. On the one hand, there are $\binom{n}{k}$ subsets containing k elements. Since k ranges from 0 to n , the number of subsets is

$$\sum_{k=0}^n \binom{n}{k}.$$

On the other hand, we can form a subset by either choosing an element or not. There are 2 choices for each element, and so there are 2^n choices in total. This establishes the identity.

21. We count the number of subsets of $\{1, 2, \dots, n+1\}$ consisting of $m+1$ elements. It is obvious that there are $\binom{n+1}{m+1}$ such subsets.

On the other hand, let $k+1$ be the largest element in the subset. Then $m \leq k \leq n$ since there are $m+1$ elements in the subset. For each fixed k , we need to choose m more elements from $\{1, 2, \dots, k\}$. There are $\binom{k}{m}$ ways to do so. This shows there are

$$\sum_{k=m}^n \binom{k}{m}$$

subsets. This establishes the identity.

22. We partition $m+n$ objects into two groups, one with m objects and the other with n objects. To choose k objects from $m+n$ objects, clearly, there are $\binom{m+n}{k}$ choices. On the other hand, we can choose j objects from the first group and hence $k-j$ objects from the second group where $j = 0, 1, \dots, k$. Hence, there are

$$\sum_{j=0}^k \binom{m}{j} \binom{n}{k-j}$$

choices in total. This establishes the identity.

23. Suppose there are n objects. We choose m of these objects and colour each of the m chosen objects in one of two colours black and white. Note that there are $\binom{n}{m}$ ways to choose the objects and 2^m ways to colour them. This gives $2^m \binom{n}{m}$ ways to finish the task.

On the other hand, we can first choose j of the objects and colour them black, where $j = 0, 1, \dots, n$. Then we choose $m-j$ of the remaining $n-j$ objects and colour them white. There are $\sum_{j=0}^n \binom{n}{j} \binom{n-j}{m-j}$ ways in total. This establishes the identity.

24. Suppose there are $2n$ people who form n pairs. There are $\binom{2n}{n}$ ways to choose n people from them. Next, we consider an alternative way to choose n people. Firstly, we partition the n pairs into 3 groups A, B, C . We are going to choose both people from each pair in group A , choose one person from each pair in group B , and do not choose any one from group C . Since n people should be chosen, we need $2|A| + |B| = n$. In particular, this shows $n - |B|$ is even.

For each $j = 0, 1, \dots, \left\lfloor \frac{n}{2} \right\rfloor$, we consider those partitions for which $|B| = n - 2j$. Then we have $|A| = \frac{n - |B|}{2} = j$ and $|A| + |C| = n - |B| = 2j$. There are $\binom{n}{2j}$ ways to choose the pairs to form groups A and C , and from these pairs there are $\binom{2j}{j}$ ways to form the group A . Also, for each of the $n - 2j$ pairs in group B , there are 2 ways to choose a person. In total, there are

$$\sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} 2^{n-2j} \binom{n}{2j} \binom{2j}{j}$$

ways to do the task. This establishes the identity.

25. Suppose there are $n + 1$ red balls labelled from 1 to $n + 1$, and n blue balls labelled from 1 to n . Let S be the set of ways to choose some balls such that the number of red balls is greater than the number of blue balls. If we choose j red balls, there are $\binom{n+1}{j}$ ways to do so. Afterwards, we may choose k blue balls where k ranges from 0 to $j - 1$. There are $\sum_{k=0}^{j-1} \binom{n}{k}$ ways to do so. Summing over j from 1 to $n + 1$, we find that $|S|$ is exactly the left-hand side of the identity.

Note that the number of ways to choose some balls is 2^{2n+1} . So it suffices to show that $|S| = |T|$ where T is the set of ways to choose the balls such that the number of red balls does not exceed the number of blue balls. We find a bijection to prove this. For each element in S , we simply change the colour of those chosen balls which have labels at most n . If the red ball with label $n + 1$ is chosen, then we remove it. Otherwise, we put it in the new set. Then we get a new choice of balls.

Suppose there are r red balls and b blue balls chosen in the old choice. If the red ball with label $n + 1$ is chosen, then there are b red balls and $r - 1$ blue balls in the new choice. As $r > b$, we have $r - 1 \geq b$. If the red ball with label $n + 1$ is not chosen, then there are $b + 1$ red balls and r blue balls in the new choice. Again, we have $r \geq b + 1$. So the new choice belongs to T . Also, if we carry out the same change again, we get back the original choice. This means the correspondence is a bijection. This shows $|S| = |T|$ and hence $|S| = \frac{1}{2} \cdot 2^{2n+1} = 2^{2n}$. This establishes the identity.

26. We count the number N of pairs $(C_1, \{J_1, J_2\})$ such that the judges J_1 and J_2 rate the same for the contestant C_1 . Firstly, for each of the $\binom{b}{2} = \frac{b(b-1)}{2}$ pairs of judges J_1 and J_2 , there are at most k contestants C_1 for which the corresponding pair is counted. So there are at most $\frac{kb(b-1)}{2}$ pairs.

Secondly, for each of the a contestants, suppose there are x judges who rate “pass”. Then there are $b - x$ judges who rate “fail”. Hence, there are exactly $\binom{x}{2} + \binom{b-x}{2}$ pairs to be counted. By the majorization inequality, as b is odd, we get

$$\binom{x}{2} + \binom{b-x}{2} \geq \binom{\frac{b+1}{2}}{2} + \binom{\frac{b-1}{2}}{2} = \frac{(b+1)(b-1)}{8} + \frac{(b-1)(b-3)}{8} = \frac{(b-1)^2}{4}.$$

As this lower bound holds for each of the a contestants, there are at least $\frac{a(b-1)^2}{4}$ pairs. It follows that

$$\frac{kb(b-1)}{2} \geq \frac{a(b-1)^2}{4}.$$

This can be simplified to give $\frac{k}{a} \geq \frac{b-1}{2b}$ as desired.

27. Let $|A| = m$. For each element $t = 1, 2, \dots, m$, let b_t be the number of subsets A_j which contain t . Then there are $\binom{b_t}{2}$ pairs of subsets A_i and A_j both containing t . This implies

$$\sum_{t=1}^m \binom{b_t}{2} = \sum_{1 \leq i < j \leq n+1} |A_i \cap A_j| \leq k \binom{n+1}{2}.$$

By Jensen’s theorem, we find that

$$\sum_{t=1}^m \binom{b_t}{2} \geq m \binom{\frac{b_1+b_2+\dots+b_m}{m}}{2} = \frac{n(n+1)(n(n+1)-m)}{2m}$$

since $b_1 + b_2 + \dots + b_m = |A_1| + |A_2| + \dots + |A_{n+1}| = n(n+1)$ by a simple double counting. It follows that $\frac{kn(n+1)}{2} \geq \frac{n(n+1)(n^2+n-m)}{2m}$. This can be simplified to $km \geq n^2 + n - m$, i.e.

$$|A| = m \geq \frac{n(n+1)}{k+1}.$$

28. For the j th row, suppose there are a_j boys and hence $75 - a_j$ girls. Let N be the number of pairs of students in the same row who have the same gender. Since there are $\binom{a_j}{2} + \binom{75-a_j}{2}$ such pairs in the j th row, we get

$$N = \sum_{j=1}^{22} \left(\binom{a_j}{2} + \binom{75-a_j}{2} \right).$$

From the condition, there are at most 11 such pairs for any pair of columns. So

$$N \leq 11 \binom{75}{2}.$$

Combining these, we find that

$$\sum_{j=1}^{22} \left(\binom{a_j}{2} + \binom{75-a_j}{2} \right) \leq 11 \binom{75}{2}.$$

By Jensen's inequality, we have

$$\binom{a_1}{2} + \binom{a_2}{2} + \cdots + \binom{a_{22}}{2} \geq 22 \binom{\frac{a_1+a_2+\cdots+a_{22}}{22}}{2}$$

and

$$\binom{75-a_1}{2} + \binom{75-a_2}{2} + \cdots + \binom{75-a_{22}}{2} \geq 22 \binom{\frac{(75-a_1)+(75-a_2)+\cdots+(75-a_{22})}{22}}{2}.$$

Let $s = a_1 + a_2 + \cdots + a_{22}$. Then we have

$$22 \binom{\frac{s}{22}}{2} + 22 \binom{75-\frac{s}{22}}{2} \leq 11 \binom{75}{2}.$$

This can be simplified to give $s^2 - 1650s + 671550 \leq 0$. Thus,

$$s \leq 825 + \sqrt{9075} < 928.$$

29. Let N be the number of pairs $(P, \{C_1, C_2\})$ such that the person P belongs to the distinct committees C_1 and C_2 . Let a_j be the number of committees that the delegate P_j joins. Note that

$$a_1 + a_2 + \cdots + a_{1600} = 16000 \times 80.$$

For each P_j , there are $\binom{a_j}{2}$ pairs for which P_j is the first entry. This gives

$$N = \sum_{j=1}^{1600} \binom{a_j}{2}.$$

By Jensen's inequality, we have

$$N \geq 1600 \binom{\frac{a_1+a_2+\cdots+a_{1600}}{1600}}{2} = 1600 \binom{800}{2} = 1600 \times 400 \times 799.$$

Therefore, by the pigeonhole principle, there exist C_1 and C_2 such that $\{C_1, C_2\}$ is the second entry of at least

$$N \div \binom{16000}{2} \geq \frac{1600 \times 400 \times 799}{8000 \times 15999} = \frac{80 \times 799}{15999} > 3$$

pairs. In other words, these two committees have at least 4 members in common.

30. (a) Suppose there is no such rectangle. Let there be b_k black squares and w_k white squares respectively in the k th column (where $k = 1, 2, \dots, 7$). Note that

$$b_k + w_k = 4$$

for each k . We count the total number N of pairs of squares in the same column having the same colour. Firstly, there are $\binom{b_k}{2} + \binom{w_k}{2}$ pairs in the k th column. By checking $b_k = 0, 1, 2, 3, 4$, we find that

$$\binom{b_k}{2} + \binom{w_k}{2} \geq 2.$$

This yields

$$N \geq 2 \times 7 = 14. \quad (1)$$

Secondly. For any pair of rows, there can be at most one pair of squares in the same column which are both black in colour. Otherwise, we can find two rows and two columns which intersect at four squares of black colour. The same holds for the white colour. Thus, there are at most 2 such pairs of squares in any pair of rows. This implies

$$N \leq 2 \binom{4}{2} = 12. \quad (2)$$

Note that (1) and (2) contradict each other. Therefore, there must be a rectangle whose four corner squares are of the same colour.

- (b) The following gives an example (B for black and W for white).

B	B	B	W	W	W
B	W	W	B	B	W
W	B	W	B	W	B
W	W	B	W	B	B

31. Let the colours be A, B, C . Let there be a_k, b_k, c_k squares of colours A, B, C respectively in the k th row (where $k = 1, 2, \dots, n+1$). Note that

$$a_k + b_k + c_k = n - 1$$

for each k . We count the total number N of pairs of squares in the same row having the same colour. Firstly, there are

$$\binom{a_k}{2} + \binom{b_k}{2} + \binom{c_k}{2}$$

pairs in the k th row. This yields

$$N = \sum_{k=1}^{n+1} \left(\binom{a_k}{2} + \binom{b_k}{2} + \binom{c_k}{2} \right). \quad (1)$$

Secondly. For any pair of columns, there can be at most one pair of squares in the same row having the same colour A . Otherwise, we can find two rows and two columns which intersect at four squares of colour A . The same holds for the colours B and C . Thus, there are at most 3 such pairs of squares in any pair of columns. This implies

$$N \leq 3 \binom{n-1}{2}. \quad (2)$$

Combining (1) and (2), we obtain

$$\sum_{k=1}^{n+1} \left(\binom{a_k}{2} + \binom{b_k}{2} + \binom{c_k}{2} \right) \leq 3 \binom{n-1}{2}.$$

By Jensen's inequality, we have

$$\begin{aligned} \binom{a_k}{2} + \binom{b_k}{2} + \binom{c_k}{2} &\geq 3 \binom{\frac{a_k+b_k+c_k}{3}}{2} = 3 \binom{\frac{n-1}{3}}{2} \\ &= 3 \cdot \frac{\frac{n-1}{3} \left(\frac{n-1}{3} - 1 \right)}{2} = \frac{(n-1)(n-4)}{6}. \end{aligned}$$

It follows that

$$\frac{(n+1)(n-1)(n-4)}{6} \leq \frac{3(n-1)(n-2)}{2}.$$

After expansion, this becomes $(n-1)(n^2 - 12n + 14) \leq 0$. As $n > 1$, we get $n \leq 6 + \sqrt{22}$. Hence, n cannot exceed 10. It is possible to have $n = 10$. The following gives one possible colouring.

A	A	A	B	B	B	C	C	C
B	B	B	C	C	C	A	A	A
C	C	C	A	A	A	B	B	B
A	B	C	A	B	C	A	B	C
B	C	A	B	C	A	B	C	A
C	A	B	C	A	B	C	A	B
C	A	B	B	C	A	A	B	C
A	B	C	C	A	B	B	C	A
B	C	A	A	B	C	C	A	B
B	C	A	C	A	B	A	B	C
C	A	B	A	B	C	B	C	A

Thus, the greatest possible value of n is 10.

32. Suppose there are $a_1, a_2, \dots, a_{2n}$ rooks in the $2n$ rows respectively. Then there are $\sum_{k=1}^{2n} \binom{a_k}{2}$ pairs of rooks lying in the same row. If n is odd, since

$$a_1 + a_2 + \dots + a_{2n} = n(n+1) + 1 = (2n-1) \cdot \frac{n+1}{2} + \frac{n+3}{2},$$

by the majorization inequality, we have

$$\begin{aligned} \sum_{k=1}^{2n} \binom{a_k}{2} &\geq (2n-1) \binom{\frac{n+1}{2}}{2} + \binom{\frac{n+3}{2}}{2} \\ &= \frac{(2n-1)(n+1)(n-1)}{8} + \frac{(n+3)(n+1)}{8} \\ &= \frac{(n+1)(n^2 - n + 2)}{4}. \end{aligned}$$

By symmetry, there are at least $\frac{(n+1)(n^2 - n + 2)}{4}$ pairs of rooks lying in the same column. Similarly, if n is even, by noting

$$a_1 + a_2 + \dots + a_{2n} = n(n+1) + 1 = (n-1) \cdot \frac{n}{2} + (n+1) \cdot \frac{n+2}{2},$$

we have

$$\begin{aligned} \sum_{k=1}^{2n} \binom{a_k}{2} &\geq (n-1) \binom{\frac{n}{2}}{2} + (n+1) \binom{\frac{n+2}{2}}{2} \\ &= \frac{(n-1)n(n-2)}{8} + \frac{(n+1)(n+2)n}{8} \\ &= \frac{n(n^2 + 2)}{4}. \end{aligned}$$

Let $b_1, b_2, \dots, b_s$ be the number of rooks that threaten each of the rooks respectively, where $s = n(n+1) + 1$. If n is odd, by a simple double counting, since each pair of threatened relations contributes 2, we get

$$\begin{aligned} b_1 + b_2 + \dots + b_s &\geq 2 \left(\frac{(n+1)(n^2 - n + 2)}{4} + \frac{(n+1)(n^2 - n + 2)}{4} \right) \\ &= (n+1)(n^2 - n + 2). \end{aligned}$$

By the pigeonhole principle, there exists b_j such that

$$b_j \geq \left\lceil \frac{(n+1)(n^2 - n + 2)}{s} \right\rceil = n - 1 + \left\lceil \frac{n+3}{n^2 + n + 1} \right\rceil \geq n.$$

This corresponds to a rook that we need. Similarly, if n is even, we have

$$b_1 + b_2 + \cdots + b_s \geq n(n^2 + 2).$$

There exists b_j such that

$$b_j \geq \left\lceil \frac{n(n^2 + 2)}{s} \right\rceil = n - 1 + \left\lceil \frac{2n + 1}{n^2 + n + 1} \right\rceil \geq n.$$

Again, we can find a rook threatened by n rooks.